

1. Credit Card Debt for Graduate Students

```
In [1]: mu = 3262  
sigma = 1100
```

1(a) A randomly selected graduate student owes at least \$1000.

```
In [2]: from scipy.stats import norm  
  
p_at_least_1000 = 1 - norm.cdf(1000, loc=mu, scale=sigma)  
print("P(X >= 1000) =", p_at_least_1000)
```

$P(X \geq 1000) = 0.9801262606889721$

1(b) A randomly selected graduate student owes more than \$4000.

```
In [3]: p_more_than_4000 = 1 - norm.cdf(4000, loc=mu, scale=sigma)  
print("P(X > 4000) =", p_more_than_4000)
```

$P(X > 4000) = 0.25113922232771857$

1(c) A randomly selected graduate student owes between 3000 and 4000.

```
In [4]: p_between_3000_4000 = norm.cdf(4000, loc=mu, scale=sigma) - norm.cdf(3000, loc=mu, scale=sigma)  
print("P(3000 <= X <= 4000) =", p_between_3000_4000)
```

$P(3000 \leq X \leq 4000) = 0.3429907384888178$

2. Charitable Contributions in Virginia, Find the limits for the middle 50% of contributions.

```
In [5]: mean=792  
sigma=103  
  
q1 = norm.ppf(0.25, loc=mu, scale=sigma)  
q3 = norm.ppf(0.75, loc=mu, scale=sigma)
```

```
print("Middle 50% range: from", q1, "to", q3)
```

Middle 50% range: from 3192.5275557298037 to 3331.4724442701963

3. Competency Test Scores

```
In [6]: mu = 400  
sigma = 100
```

3(a) What is the minimum score needed to be in the top 3% of students?

```
In [7]: score_top_3_percent = norm.ppf(0.97, loc=mu, scale=sigma)  
print("Minimum score for top 3%:", score_top_3_percent)
```

Minimum score for top 3%: 588.079360815125

3(b) What is the minimum score needed to avoid being in the bottom 1.5%?

```
In [8]: score_bottom_1_5_percent = norm.ppf(0.015, loc=mu, scale=sigma)  
print("Minimum score to avoid summer school:", score_bottom_1_5_percent)
```

Minimum score to avoid summer school: 182.99096224154394

4. Transistor Lifetime (Gamma Distribution)

```
In [9]: from scipy.stats import gamma  
  
mean = 10  
variance = 50  
  
# x * theta = 10  
# x * theta **2 = 50  
  
alpha = 2  
beta = 0.2  
theta = 1 / beta
```

4(a) Find the parameters α and β of the gamma distribution.

```
In [10]: # alpha * theta = 10
# alpha * theta **2 = 50
# 10 / alpha**2 * alpha = 50

alpha = 2
beta = 0.2
theta = 1 / beta
```

4(b) What is the probability that a transistor lasts at most 50 weeks?

```
In [11]: p_leq_50 = gamma.cdf(50, a=alpha, scale=theta)
print("P(X <= 50) =", p_leq_50)
```

$P(X \leq 50) = 0.9995006007726127$

4(c) What is the probability that a transistor does not survive the first 10 weeks?

```
In [12]: p_leq_10 = gamma.cdf(10, a=alpha, scale=theta)
print("P(X <= 10) =", p_leq_10)
```

$P(X \leq 10) = 0.5939941502901616$

5. Committee Selection from Foreign Student Club

```
In [13]: from math import comb

canadians = 2
japanese = 3
italians = 5
germans = 2

total_people = canadians + japanese + italians + germans
total_committees = comb(total_people, 4)
```

5(a) Find the probability that all nationalities are represented in the committee.

```
In [14]: ways_all_represented = comb(canadians, 1) * comb(japanese, 1) * comb(italians, 1) * comb(germans, 1)
p_all_represented = ways_all_represented / total_committees
p_all_represented
```

Out[14]: 0.12121212121212122

5(b) Find the probability that all nationalities except Italian are represented.

```
In [15]: ways_non_italians_valid = (  
    comb(canadians, 2) * comb(japanese, 1) * comb(germans, 1)  
    + comb(canadians, 1) * comb(japanese, 2) * comb(germans, 1)  
    + comb(canadians, 1) * comb(japanese, 1) * comb(germans, 2)  
    )  
    p_no_italians = ways_non_italians_valid / total_committees  
    p_no_italians
```

Out[15]: 0.048484848484848485

6. Light Bulb Lifespan Probability; Find the probability of exactly four bulbs lasting less than 500 hours and two lasting more than 800 hours in a sample of ten.

```
In [16]: from math import factorial  
  
p_A = 0.5  
p_B = 0.2  
p_C = 0.3  
  
n = 10  
a, b, c = 4, 2, 4  
  
coefficient = factorial(n) // (factorial(a) * factorial(b) * factorial(c))  
probability = coefficient * (p_A ** a) * (p_B ** b) * (p_C ** c)  
  
print(f"Probability = {probability}")
```

Probability = 0.06378750000000001